

Comparative Analysis of Heart Failure Detection via Machine Learning

Sowndaryan Jayaprakashanand (*Author*)
Sowndaryan_Jayaprakashanand@student.uml.edu

Mohammed Aldulaimy (*Author*)
Mohammed_Aldulaimy@student.uml.edu

Lucas Brandao (*Author*)
Lucas_Brando@student.uml.edu

Miraaj Chaudhry (*Author*)
Miraaj_Chaudhry@student.uml.edu

Abstract—Neural Networks are frequently adopted in clinical prediction tasks due to their ability to model complex, non-linear relationships in data. However, their advantage over classical models is less established on small, tabular datasets like ones used in clinical settings. This project aims to compare the performance of a Neural Network against classical machine learning models (Random Forest, Logistic Regression, and Support Vector Classifier) on a real-world dataset of 299 heart failure patients, with death during follow-up as the prediction target. Results show classical models consistently outperforming the Neural Network model with Random Forest achieving the lowest false negative rate, a critical metric in the clinical field. These findings suggest that model complexity does not guarantee improved performance at this scale. Future work includes repeated evaluation on larger datasets, improved class imbalance handling, and Neural Network architecture optimization.

Keywords— *Heart failure; neural networks; decision trees; Random Forest; support vector classifier; logistic regression; recall;*

I. INTRODUCTION

Cardiovascular diseases are the leading cause of death worldwide [1], contributing to ~681,000 deaths annually in the United States alone. Heart failure is a common condition caused by cardiovascular diseases, and early detection is necessary for preventing deaths from this health issue.

Traditionally, clinicians evaluate the risk of cardiovascular disease using methods like Logistic Regression models or Framingham Risk Scores [2]. These methods, while interpretable, often struggle to capture complex, non-linear interactions found in clinical tabular data, such as correlations between physiological features and lifestyle factors. Random Forest ensemble models and Neural Networks have thus been adopted in the field, aiming to improve prognostic accuracy by better defining these complex correlations.

Despite growing adoption of machine learning in clinical settings, it remains unclear whether neural networks offer

meaningful advantages over classical models on small, tabular datasets typical of clinical records. Support Vector Machine models have gained traction in the clinical research industry for tasks such as prognosis, and prediction of disease outcomes [3], similarly with Random Forest ensemble models [4]. The performance of Logistic Regression, Random Forest, and Support Vector Machine models will be compared against a Neural Network on a 299 patient dataset titled “Heart Failure Clinical Records” from the UC Irvine Machine Learning Repository [5].

Our results indicate that classical models (particularly Random Forest) consistently outperform the Neural Network on this dataset, suggesting that model complexity does not guarantee performance gains at this scale. The remainder of this paper is organized as follows: Section II covers background / related work, Section III describes the dataset, Section IV details our methodology, Section V presents evaluations from results, and Section VI concludes with aims for future work.

II. BACKGROUND / RELATED WORK

The application of machine learning in cardiovascular disease prognosis represents a significant evolution from traditional clinical risk assessments. Historically, clinicians have relied on tools such as Framingham Risk Scores or standard logistic regression paradigms to evaluate patient risks. While highly interpretable, these classical methods often struggle to capture the complex, non-linear interactions found in physiological and lifestyle data. As a result, more sophisticated algorithms have gained traction in clinical research. For instance, Support Vector Machine (SVM) models and Random Forest ensembles have been increasingly utilized for disease prognosis and outcome prediction due to their robustness and predictive power. The specific dataset utilized in this study, the Heart Failure Clinical Records dataset, was initially compiled and analyzed by Ahmad et al.,

who conducted a baseline survival analysis on the 299-patient cohort from the Faisalabad Institute of Cardiology and Allied Hospital. Following its publication, the dataset has served as an important benchmark for evaluating clinical predictive models. Chicco and Jurman conducted an extensive analysis on this exact cohort, demonstrating that machine learning can effectively predict patient survival using a subset of clinical features [6]. Through repeated evaluations across various algorithms, their results indicated that a Random Forest classifier yielded the most reliable performance metrics [6]. Despite the established success of these ensemble methods, Neural Networks are frequently championed as the definitive state-of-the-art solution for complex predictive modeling tasks. However, their purported advantage over classical machine learning models remains debated when applied to the small, tabular datasets typical of everyday clinical records. Deep learning models inherently require vast amounts of data to optimize complex architectures and avoid overfitting, a luxury rarely afforded in specialized, real-world clinical cohorts. This comparative analysis builds directly upon the foundation laid by Ahmad et al. and Chicco and Jurman by explicitly contrasting the established efficacy of classical models against a multi-layer Neural Network. By doing so, we aim to investigate whether the theoretical complexity of deep learning translates into tangible clinical benefits. Specifically regarding the critical reduction of false negatives, when constrained by a realistically small medical dataset.

III. DATASET

The dataset used in this project is the “Heart Failure Clinical Records” dataset, sourced from the UC Irvine Machine Learning Repository [5]. Originally collected by Tanvir Ahmad et al. at the Faisalabad Institute of Cardiology and Allied Hospital in Faisalabad, Pakistan, between April-December 2015 [7], it contains medical records of 299 patients who had left ventricular systolic dysfunction and had previous heart failures that put them in classes III or IV or New York Heart Association (NYHA) classification of the stages of heart failure. The patient cohort consists of 194 men and 105 women ranging in age from 40-95 years.

The dataset contains 13 clinical features per patient, summarized in Table I.

Feature	Type	Unit	Description
age	Integer	Years	Age of the patient
anaemia	Binary	—	Decrease of red blood cells or hemoglobin
creatinine phosphokinase	Integer	mcg/L	Level of the CPK enzyme in the blood
diabetes	Binary	—	Whether the patient has diabetes
ejection fraction	integer	%	Percentage of blood leaving the heart at each contraction
high blood pressure	binary	—	Whether the patient has hypertension
platelets	Continuous	kiloplatelets/mL	Platelets in the blood
serum creatinine	Continuous	mg/dL	Level of serum creatinine in the blood
serum sodium	Integer	mEq/L	Level of serum sodium in the blood
sex	Binary	—	Sex of the patient (male/female)
smoking	Binary	—	Whether the patient smokes
time*	Integer	Days	Follow-up period (excluded – data leakage)
DEATH_EVENT	Binary	—	Target: whether patient died during follow-up

Table I. Clinical Features of the Heart Failure Clinical Records dataset [5]. The time feature was excluded from all models due to data leakage.

The features of this dataset fall into two categories: six continuous measurements and five binary indicators. Continuous features include age, creatinine phosphokinase, ejection fraction, platelets, serum creatinine, serum sodium. Binary indicators include anaemia, diabetes, high blood pressure, sex, and smoking status. The target variable, DEATH_EVENT, is a binary indicator of whether or not the patient died during the follow up period.

IV. EXPLORATORY DATA ANALYSIS AND PREPROCESSING

Exploratory data analysis was conducted to understand the distribution and relationships within the dataset. The dataset contains no missing or null values across all patient records, and all binary features were encoded as 0 or 1, requiring no additional encoding. Continuous features are largely uncorrelated with one another through examination of the

feature correlation heatmap. Several continuous features exhibit significant right skew. Serum creatinine and platelets had a skewness of 4.46 and 1.46 respectively, while serum sodium was mildly left-skewed at -1.05. This motivated the application of a Box-Cox transformation during preprocessing.

The target variable distribution confirmed a class imbalance in the dataset: approximately 67.9% of patients survived and 32.1% died during follow-up. This imbalance informed subsequent modeling decisions including class weighting and metric selection.

Prior to data preprocessing, the time feature was explicitly excluded. This feature records the length of each patient's follow-up period in days. Because patients who died were monitored for a shorter duration, time becomes a direct proxy for the prediction target and constitutes data leakage. Including this feature in real clinical settings where follow-up duration is unknown at the time of prediction is unrealistic, and was thus removed during data-processing.

The dataset was partitioned into three non-overlapping subsets: 60% training, 20% validation, and 20% test. Stratified splitting was applied at both stages using scikit-learn's [8] 'train_test_split' with 'stratify=y', ensuring a ~32% positive class ratio across all three subsets. The validation set was reserved primarily for neural-network training monitoring. All classical models were tuned exclusively using cross-validation on the training set and evaluated once on the held-out test set.

Continuous and binary features were preprocessed separately using a 'ColumnTransformer' pipeline. Continuous features were first transformed using a Box-Cox transformation via scikit-learn's 'PowerTransformer', which reduces right skew and stabilizes variance by finding the optimal power parameter λ for each feature. Box-Cox requires all input values to be strictly positive, which was the case for all continuous features in the dataset. The transformed features were then standardized using 'StandardScaler', which centers each feature to zero mean and unit variance. Binary features were passed through the pipeline unchanged.

Critically, the preprocessing pipeline was fit only on the training set and applied to the validation and test sets using the learned parameters. Fitting the scaler on the full dataset would allow models to indirectly observe the distribution of held-out data. Doing so would introduce data leakage, and violate the integrity of the evaluation.

V. METHODOLOGY

All models were evaluated on a common held-out test set using precision, recall, F1-score, and confusion matrices [9]. Given the clinical context of this research, recall on the Deceased class was treated as the primary metric of interest, as false negatives (patients who died but were predicted to survive) represent the most dangerous failure type. Accuracy was not prioritized due to the class imbalance; a naive

classifier predicting survival for every patient would achieve 67.9% accuracy while detecting zero deaths. All hyperparameter tuning was performed exclusively on the training set via cross-validation, with the test set evaluated only once per model. The specific parameters to tune were chosen through rigorous testing, starting with parameters known to help each model's performance with medical data [10].

The first model implemented was a logistic regression model using scikit-learn. GridSearchCV was utilized with 10-fold-cross-validation to find the best combination of parameters for each of the following metrics: recall, F1, and precision. The parameters for this model included C-values, which help the model avoid overfitting, of 0.01, 0.1, 1.0, 10.0, and 100.0. Class weights of 'balanced' and 'None' were also added as there was an imbalance in the target class, and all six solvers available were tested. Both the optimal F1 and optimal recall models had the same parameters of a C-value of 1.0, a solver of 'liblinear', and class weights being 'balanced'. These parameter settings differ a bit from the default settings of the solver being 'lbfgs' and class weights being 'None', while the C-value stayed at its default of 1.0. To analyze the trade-offs between precision and recall, the model was modified with a lower threshold to better understand its performance.

A support vector classifier (SVC) was implemented using scikit-learn's [7] SVC estimator. Hyperparameter tuning was performed using GridSearchCV with 10-fold cross-validation across recall, precision, and F1 objectives. The parameter grid covered a range of C values $\in \{0.01, 0.1, 1.0, 10.0, 100.0\}$, kernel type $\in \{\text{linear}, \text{rbf}\}$, gamma $\in \{\text{scale}, \text{auto}\}$, and both balanced and unbalanced class weighting. The F1-tuned model was selected as the final model as it produces the most clinically balanced tradeoff between precision and recall rather than optimizing either in isolation.

A Random Forest classifier was implemented using scikit-learn's RandomForestClassifier. To address the class imbalance, models were evaluated both with 'balanced' class weighting. An iteration of random undersampling was attempted but quickly discarded after understanding that the initial dataset only consisted of 299 patients. Hyperparameter optimization was conducted using GridSearchCV with 5-fold cross-validation, focusing on the recall metric to minimize false negatives. The search space included the number of estimators $\in \{50, 100, 200, 300\}$, maximum tree depth $\in \{\text{None}, 5, 10, 20\}$, and the minimum samples required to split a node $\in \{2, 5, 10\}$. The final model configuration favored a balanced class weight and a minimum split requirement of 10, achieving a more robust performance on the deceased class compared to the baseline linear models.

A Multi-Layer Perceptron (MLP) neural network was designed to evaluate deep learning performance on the clinical dataset. The architecture consisted of an input layer scaled to the preprocessed clinical features, followed by two hidden

layers. The first hidden layer was configured with 32 neurons, and the second with 16 neurons. Both hidden layers utilized the Rectified Linear Unit (ReLU) activation function to model complex, non-linear interactions within the data. To mitigate the risk of overfitting on the relatively small dataset, dropout regularization was applied subsequent to both hidden layers. The output layer consisted of a single neuron using a sigmoid activation function to yield a probability score for the binary prediction target (DEATH_EVENT). The network was compiled using Binary Cross-Entropy (BCE) as the objective loss function and optimized with a learning rate of 0.001. Model training was configured for a maximum of 150 epochs utilizing a batch size of 32. To further guard against overfitting and ensure optimal weight retention, an early stopping callback was implemented monitoring the validation loss with a patience of 20 epochs. Furthermore, to directly address the established class imbalance, two distinct variants of the neural network were trained: an unweighted baseline model and a class-weighted model. The class-weighted variant assigned higher penalty weights to the minority (deceased) class during loss calculation, strategically aiming to improve the critical recall metric.

VI. RESULTS

As mentioned earlier, the metrics used to evaluate these models included recall, precision, and F1 scores, with an emphasis placed on optimizing recall scores [9]. We also utilized confusion matrices to analyze the count of false negatives. Our initial neural network was unweighted with a recall of 0.42 in death cases and 0.83 in survivors cases, with their precision scores being 0.53 and 0.76, respectively. This model had 34 true negatives, but only 8 true positives as 11 false negatives were observed. As less than half the true death cases were captured, we introduced a weighted model and reduced the threshold in hopes to increase the recall score. The results of this weighted model can be found below in table II. The recall score in the death class increased to 0.95, but reduced to 0 in the survivors class. This was seen as the number of false negatives dramatically dropped to near 0. However, the number of true negatives dramatically dropped to near 0 as well, leading to a drop in precision in both target classes. Ultimately, this model proved unreliable in this scenario as it was predicting all patients would die. This can lead to panic, fear, and improper care of patients if applied in the real-world.

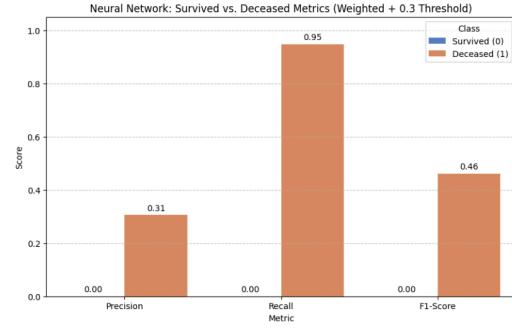


Table II Results from weighted neural network model

Our logistic regression model produced more reliable metrics. Our initial model scored a recall of 0.68 in survivors and 0.53 in deceased cases. Precision scores were 0.76 and 0.43, respectively. The model predicted 10 of the 19 actual deaths, showing the need to increase recall once again. Again, the threshold was lowered, this time to 0.35. This model's results can be seen in table III as recall shot up to 0.68 in the death class, but precision fell to 0.42. The model lost a few true negatives, but lowered its false negative from 9 to 6. Its F1 score rose from 0.48 to 0.52, showcasing its overall improvement in performance. Lowering the threshold even more resulted in the recall increasing and false negatives dropping, but the number of true negatives started to dramatically fall as well. This in turn showed 0.35 was the "sweet spot" for the threshold as it produced the best overall results for the model. The model at a 0.35 threshold still proved more reliable than the weighted MLP because even with a lower threshold it accurately predicted more survivors than the neural network and scored better precision and f1 scores.

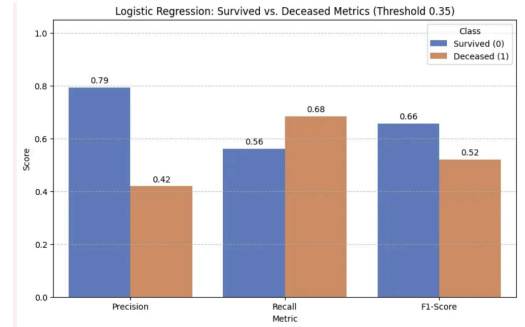


Table III Results from Logistic Regression model

The SVC produced similar results as the initial logistic regression model. As seen in table IV, precision and recall scores were 0.45 and 0.53 in the death class; 0.76 and 0.71 in the survivors class. Again, the model predicted only 10 out of 19 actual death cases, but it captured most of the surviving class and scored even higher F1 scores than the weighted MLP and regression model at a 0.35 threshold. However, its recall remains low, deeming it a well performing model, with room for improvement.

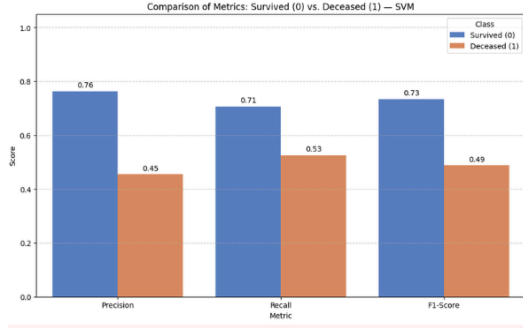


Table IV Results from Support Vector Classifier model

The third and final model was a Random Forest classifier, whose metric results can be seen in table V. As seen in the table, the model scored high in recall in both classes, with only the weighted MLP scoring a higher recall (which was due to the MLP predicting all deaths). In addition to the high recall, its precision and F1 scores in the death class were the highest across all models, as were its scores in the survivors class across all models. This indicates the model performed the best overall and this statement is only strengthened once its confusion matrix results are analyzed. The model predicted only 6 false negatives, which was the lowest across all models, excluding the weight MLP predicting all deaths. This low score with the highest count of true negatives and true positives, proved the Random Forest model to be the most reliable and best performing in our study.

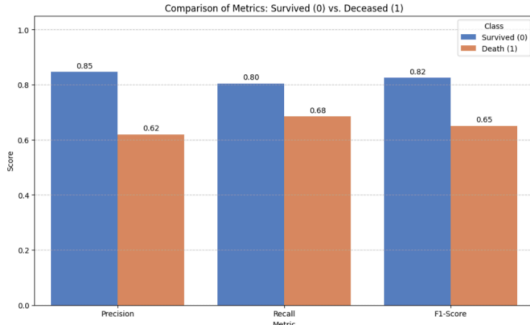


Table V Results from Random Forest model

Overall, the classical methods were able to increase their recall scores, not as much as the MLP, but their precision-recall trade-offs weren't as significant. This is visible in the tables as the classical models maintained higher precision, and F1, scores as compared to the neural network. Across all these models, and their corresponding bar charts, One class scored higher than the other in each metric. This is a direct response to their being a slight imbalance in the target class within our dataset. The precision-recall trade-off can also be seen as optimizing recall scores, which decreases models' overall precision. The only exception was the Random Forest which maintained a high score across all metrics, illustrating the advantage of the ensemble method in the medical field.

VII. CONCLUSION / FUTURE WORK

From the performance of our models, we can conclude that the classical models, specifically the Random Forest, provide an advantage over a state-of-the-art neural network. The neural network was deemed unreliable as it defaulted to labeling patients as deceased, which can cause panic, fear, and deaths if implemented in the real-world. The Random Forest was deemed more reliable than the logistic regression and SVC model as it produced a higher number of true positives and lower false negatives. Our logistic regression and SVC models missed roughly half the cases of deceased patients, showing their lack of reliability. The Random Forest's recall of **0.68** for deceased patients outperformed the scores of the SVC and logistic regression. While the regression and Support Vectors models may achieve a recall of 0.68 or close to 0.68 when their thresholds are lowered, our Random Forest achieved much higher scores in precision and F1 using default thresholds, thus making it a better model overall. Unlike the neural network which defaulted to labeling patients as dead as the threshold decreased, the classical models not only lowered false negatives, but kept a relatively high amount of true negatives as their thresholds decreased. The classical models' results would be trusted more than the neural networks by doctors and patients alike.

Our findings support research conducted by Davide Chicco and Giuseppe Jurman on the same dataset. As seen below in table VI, after applying their models 100 times each, they found a Random Forest that performed the best [8][7] and scored similar metrics as our model, indicating we've taken steps in the correct direction.

Method	MCC	F1 score	Accuracy	TP rate	TN rate	PR AUC	ROC AUC
Random forests	+0.384*	0.547	0.740*	0.491	0.864	0.657	0.800*
Decision tree	+0.376	0.554*	0.737	0.532*	0.831	0.506	0.681
Gradient boosting	+0.367	0.527	0.738	0.477	0.860	0.594	0.754
Linear regression	+0.332	0.475	0.730	0.394	0.892	0.495	0.643
One rule	+0.319	0.465	0.729	0.383	0.892	0.482	0.637
Artificial neural network	+0.262	0.483	0.680	0.428	0.815	0.750*	0.559
Naive bayes	+0.224	0.364	0.696	0.279	0.898	0.437	0.589
SVM radial	+0.159	0.182	0.690	0.122	0.967	0.587	0.749
SVM linear	+0.107	0.115	0.684	0.072	0.981*	0.594	0.754
k-nearest neighbors	-0.025	0.148	0.624	0.121	0.866	0.323	0.493

Table VI. Chicco and Jurman's models' performance metrics, averaged after 100 runs [8]

From their research and our findings, we can conclude more work can be done to find the most optimal model to help both heart failure patients and their doctors in the real-world. Moving forward, we hope to experiment with the researcher's top performing models to determine if we can optimize them, as well as potentially combine different models through ensemble methods to help combat the precision-recall trade-off. We can also rigorously explore different network architectures to see if our neural network can give more reliable results.

Furthermore, we will explore different sampling techniques to see if we can generate models with more balanced results in both the death and survived target classes. We will also explore ways to gain more data as our current dataset of 299 patients is relatively small, which may be harming our models' performances. Introducing a larger dataset may improve the results of our models, while better preparing them for real-world implications. Different methods of choosing hyperparameter values can be explored as several exist, each with its own limitations [9].

Neural networks have been seen as the “go-to” model in the medical community, but our research shows classical models can not only compete with, but can outperform deep learning in real-world applications. These findings have opened the door for more research to rigorously study how different machine learning models and techniques perform in a medical setting to deliver proper care to patients and potentially save lives.

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